IT Italiano · GB **English**

🎮 ichosynth — Usage Manual

How to play the sampler-sequencer you *draw*

You draw music on a 16×16 grid with **4 knobs** and **3 buttons**. No computer, no menus to memorize: turn, press, listen.

Level Beginner | Build 4 encoders + 3 buttons | firmware TŒRN · SP_ | See also Build Manual

> 🎧 **You don't need a computer to play**: your **ichosynth** generates everything on its own. Plug in your **headphones**, power it via USB, and off you go.

> 🔌 **Connections — getting sound out (and in)**: the instrument has three audio jacks.
>
> - **Line Out** — 2× **6.35 mm (1/4") MONO** jacks, **L + R = stereo**. Plug them into an amplifier, mixer, PA or audio interface. The output is stereo (TŒRN pans the voices), so use **both L and R**; for a mono rig just use the **L** jack.
> - **Line In** — 1× **6.35 mm (1/4") MONO** jack. Feed in an instrument, another synth, a field recorder or a mixer send to **record/sample it** (see ch. 16).
> - **Headphone** — the on-board **3.5 mm stereo** jack, for monitoring.
>
> All three jacks share the device's ground, and **Line Out and headphone play at the same time** — no menu setup needed.

> ℹ️ **This is the real TŒRN.** ichosynth runs the full **TŒRN firmware** (by SP_/soundpauli, toern.live) on a Teensy 4.1, built with cheap hand-soldered parts: **4 KY-040 encoders**, **3 tact switches** and an **SSD1306 OLED**. Where the original TŒRN told you the state with the glowing colour of its encoder rings, our build shows it as plain text on the **OLED**.

📑 Table of Contents

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1 · The concept in 30 seconds

The **16×16 grid** is your music sheet. A playback "playhead" runs from left to right: every column it touches plays the notes you've put there.

- The **columns** (left→right) are the **16 steps** of one bar (a page can be chained to **32×16**).
- Each **row** is a **voice** (a sample or a synth), identified by a **color**.
- Several pages in a row make up a **pattern**; several patterns make up a **song**.

> 💡 Basic flow: **draw notes → press PLAY (B1) → loop**. You change samples, BPM and volume on the fly, without stopping.

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2 · The hardware: 4 knobs + 3 buttons

On top there are **4 encoders — E1 E2 E3 E4**, left to right. Below them sit **3 tact switches — B1 B2 B3**. Every encoder both **turns** and **pushes** (click).

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[ E1 ]   [ E2 ]   [ E3 ]   [ E4 ]     ← 4 knobs (turn + push)
[  B1  ] [  B2  ] [  B3  ]            ← 3 buttons
```

The gestures the firmware understands:

Gesture	What it means
Turn (E1–E4)	change the value / move the cursor in the current context
Short click (push an encoder)	confirm / act (the meaning depends on the mode)
Long press (hold an encoder)	a second action (e.g. E1 hold = mute/subpattern)
Tap a button (B1/B2/B3)	the button's main function

Gesture	What it means
Hold a button	the button's held function (e.g. B3 held = recording count-in)
Combo B1+B2	a shortcut — **the order you press them matters** (see ch. 11 & 12)

> 💡 There is **no double-click**: the KY-040 encoders don't need one. Every action is a turn, a click, a hold, a button tap or a two-button combo.

What each encoder does changes with the mode. Here is the overview (details in the relevant chapters):

Mode	E1 turn	E2 turn	E3 turn	E4 turn
DRAW	Y / note (row)	page	quick channel filter	X / column
SINGLE	channel	note (pitch)	—	X / column
FILTER	slider 1	slider 2	slider 3	slider 4
MENU	—	value	value	navigate menu pages
VELOCITY	velocity	probability	channel volume	condition / timing
SONG	—	pattern	—	position (1–64)

The most useful encoder **clicks** (in DRAW/SINGLE):

- **E3 click** = **Play / Pause**.
- **E1 long-press** = enter mute / subpattern (release to restore).
- **E1 click @ row 16** (in SINGLE) = NOTE SHIFT.
- **E4 click** = confirm / enter a sub-menu.
- **E1 click** (in MENU) = back / exit.

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3 · Reading the grid and the OLED

- **Rows** = the voices (8 sample voices + 3 synth voices), each with its own **color** (see ch. 13).
- **Columns** = the 16 steps of the current page.
- **Top row (status)**: page indicators and status lights (copy active, etc.). During Play the page indicators turn **green**.
- **Play playhead**: the highlighted column that advances as you play (the ▼ in the image above).

> 📟 The original TŒRN showed the live state through the **colour of the encoder rings**. Our build has no RGB rings, so the **OLED** shows the same information in plain text: **current channel, mode, transport (PLAY / REC / STOP), BPM, volume and page**. Glance at it any time to know exactly where you are.

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4 · The 3 buttons (B1 B2 B3)

The three tact switches are the heart of the transport and navigation:

Button	Main function	Also does...
B1 · PLAY	start playback; toggle **SINGLE**; exit other modes back to **DRAW**	**held at power-on** = clear the RAM (fresh start)
B2 · MENU	enter / exit the **Menu** and the sub-modes	exits a sub-mode back to DRAW/SINGLE
B3 · REC	**record** (tap or hold)	**PAUSE** during playback · **tap-tempo BPM** · hold **>300 ms** = recording **count-in**

> 💡 **PLAY is B1** (a single tap), not an encoder gesture. **PAUSE is B3** while a song is running.

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5 · DRAW mode (drawing)

This is the main screen, the default one, where you create patterns.

Action	Gesture
🧭 **Move the cursor**	turn **E1** (up/down, row/note) and **E4** (left/right, column)
📄 **Change page**	turn **E2**
✏️ **Add / change a note**	push on the cursor's spot (you hear it right away); pressing again on a note cycles the voice
▶️ **Play / Pause**	**E3 click** (Pause during play is also **B3**)
🎚️ **Quick filter the channel**	turn **E3** to soften/brighten the voice under the cursor on the fly
🔇 **Mute / subpattern**	**hold E1** (release to restore)

> 💡 E3 is the "transport" knob in DRAW: **click** it for Play/Pause, **turn** it to nudge the current channel's filter without leaving the grid.

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6 · Pages, patterns and subpatterns

- The grid shows **one page** (16 steps) at a time; turn **E2** to move between pages.
- Pages with notes are played in sequence on a loop to form a **pattern**. Patterns are chained in SONG mode.
- **Subpatterns**: **hold E1** to drop into a momentary variation of the current pattern, then release to snap back — great for live fills and breaks.
- **Note copy / paste and note-shift** let you move and duplicate steps; the active copy/note-shift state shows on the grid's top row (and on the OLED).

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7 · Mute (silencing voices)

- Put the cursor on a voice and **hold E1** to mute it / drop into its subpattern; release to restore.
- Muted channels are shown **on the grid itself**, so you can always see what's silent at a glance.

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8 · Volume and BPM

- **Volume** and **BPM** are always readable on the **OLED**.
- **Tap-tempo**: tap **B3 (REC)** in time to set the **BPM** by ear.
- The per-channel **volume** is adjusted in VELOCITY mode with **E3**.

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9 · Velocity, probability and conditions

TŒRN gives every step more than just on/off. In **VELOCITY** mode the four knobs become per-step controls:

Knob	Controls
E1	**velocity** (how loud the step is)
E2	**probability** (chance the step fires)
E3	**channel volume**
E4	**condition / timing** (e.g. play every Nth pass, micro-timing)

Use **B2 (MENU)** to step through the sub-modes; **B1 (PLAY)** returns you to DRAW.

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10 · SINGLE mode (one voice only)

Useful for working in detail on one sample (e.g. a melody across several pitches).

- **Enter / exit SINGLE**: tap **B1 (PLAY)** to toggle SINGLE for the focused voice.
- In SINGLE: **E1 turn** = pick the **channel**, **E2 turn** = the **note (pitch)**, **E4 turn** = the **column (X)**.
- The same drawing/deleting gestures from DRAW apply, but only to the selected voice.

Note Shift (in SINGLE)

- Put the cursor on row 16 and **click E1** to enter **NOTE SHIFT**, then move the notes with the knobs.

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11 · FILTER mode and per-voice DSP

ichosynth has the full TŒRN per-voice DSP: a **lowpass filter**, **reverb**, **bitcrusher**, **detune**, **octave**, and a **Moog ladder** on the synth voices.

- **Enter FILTER mode**: press **B2 + B1 together, with B2 first** (B2-first = FILTER MODE). This works on the channels that have filters (1–8, 11, 13–14).
- In FILTER mode the four knobs become **four sliders** (**E1 E2 E3 E4**) for the DSP of the selected voice. The OLED shows the selected filter and its value.
- Exit with **B2 (MENU)** or **B1 (PLAY)** back to DRAW.

> 💡 For a fast, no-mode tweak you can also just **turn E3 in DRAW** to filter the channel under the cursor. FILTER mode is for dialling in the full set of effects per voice.

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12 · Changing the sample (Sample Browser)

To assign a different WAV to one of the 8 sample voices:

1. **Open the Sample Browser**: press **B1 + B2 together, with B1 first** (or both at once). This enters SET_WAV for channels 1–8.
2. **Navigate** with the knobs: change **folder** and **sample** (browse the SD); you can preview the **length** and **waveform**, and set **seek / length / reverse**.
3. **Confirm** with **E4 click** to load the selected sample onto the voice.
4. **Exit** with **B2 (MENU)**.

> 📁 Samples live on the microSD as `/samples/<folder>/_<number>.wav` (see the build manual). WAVs should be mono, 16-bit, 44.1 kHz — `wavmaker.py` prepares them for you.

> ⏱️ **Order matters in the combo**: **B2 first** → FILTER MODE; **B1 first (or simultaneous)** → SAMPLE BROWSER. There is a short 350 ms cooldown between combos.

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13 · Voice colors and the synth voices

Each voice has a fixed color (defined in `colors.h`):

Legenda colori delle voci

Ogni riga della griglia è una voce, identificata da un colore (da colors.h).

Voce 1	rosso	voce campione
Voce 2	blu	voce campione
Voce 3	giallo	voce campione
Voce 4	verde	voce campione
Voce 5	magenta	voce campione
Voce 6	verde-lime	voce campione
Voce 7	arancione	voce campione
Voce 8	turchese	voce campione
Voce 13	viola	synth (sawtooth)
Voce 14	bianco	synth (square)

There are **8 sample voices** plus **3 synth voices**. The synth voices play internally generated waves and follow the pitches of a scale; they have their own **Moog ladder** filter (set in FILTER mode). TŒRN is **polyphonic**, so several voices sound together.

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14 · Sample packs

A "sample pack" is a complete set of voices saved on the SD: recall a whole kit on the fly.

- Open the Sample Browser (ch. 12) and use the pack controls to **load** a numbered pack onto the 8 sample voices.
- Packs let you swap an entire kit between songs without rebuilding it voice by voice.

> 📁 On the SD a pack is a numbered folder with its `.wav` files inside; ichosynth manages them, you don't need to create them by hand.

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15 · Saving and loading (Menu)

- **Enter the Menu**: tap **B2 (MENU)**.
- Navigate the menu pages with **E4** (turn), adjust values with **E2 / E3**, and **confirm / enter a sub-menu with E4 click**; **E1 click = back**.
- From the Menu you **save** and **load** songs to/from the microSD.
- **Exit** the Menu with **B2 (MENU)** again, or **B1 (PLAY)** to jump straight back to DRAW.

16 · Live recording (REC)

ichosynth records audio straight from its input into the current channel.

> 🔌 **The Line In jack is the default record source.** Connect an instrument, another synth, a field recorder or a mixer send to the **6.35 mm (1/4") MONO Line In** jack and you can sample it straight in — no menu setup needed (the take is a mono sample). See the **Connections** note at the top of the manual for the jacks.

1. Choose the **channel** to record into.
2. **Hold B3 (REC)** to record from the input (**MIC** or **LINE**, the default; the choice shows on the OLED).
3. **Hold B3 > 300 ms** to get a **count-in** first (4 beats at the current BPM), then recording starts on the beat.
4. Release **B3** to stop; the take is saved to the SD and loaded onto the channel, so it survives a reboot.

> 💡 **B3 is multi-purpose**: a quick tap is **tap-tempo** / **PAUSE during play**; a hold is **record**. The OLED's transport readout shows **REC** while you're recording.

17 · SONG mode

SONG mode chains your patterns into a full arrangement.

- In SONG mode: **E2 turn** = choose the **pattern**, **E4 turn** = the **position** in the song (1–64).
- Build the sequence of patterns, then **PLAY (B1)** runs the whole song.
- Mute states and the song position are shown **on the grid**, so you can follow the arrangement live.

18 · MIDI and tap-tempo

- **USB MIDI**: ichosynth is a **USB MIDI** device. Connected over USB it plays/receives MIDI with a computer or other gear.
- **Tap-tempo**: tap **B3 (REC)** in time to set the BPM by feel — no menu needed.

19 · Mode & command map

From DRAW (the main screen) you reach everything with knobs, buttons and the two combos:

Mappa delle modalità — cosa fa ogni encoder

I 4 encoder cambiano funzione col modo. I 3 pulsanti restano costanti.

	E1 ruota	E2 ruota	E3 ruota	E4 ruota
DRAW	Y / nota	pagina	filtro rapido	X / colonna
SINGLE	canale	nota (pitch)	—	X / colonna
FILTER	slider 1	slider 2	slider 3	slider 4
MENU	—	valore	valore	naviga pagine
VELOCITY	velocity	probabilità	volume canale	condizione
SONG	—	pattern	—	posizione

I 3 pulsanti (sempre attivi):

B1 PLAY
play/pausa · SINGLE · esci

B2 MENU
menu · tieni = FX MODE

B3 REC
rec · tap-tempo · tieni = count-in

Legend: **E1–E4** = the four knobs (turn or click) · **B1/B2/B3** = the three buttons · "click" = short press · "hold" = long press · "combo" = two buttons together (order matters).

You want to...	Gesture
Move cursor up/down (row/note)	turn **E1**
Move cursor left/right (column)	turn **E4**
Change page	turn **E2**
Add / change a note	push on the cursor's spot
Play / Pause	**E3 click** (Pause also **B3**)
Quick-filter the current channel	turn **E3** (in DRAW)
Mute / subpattern (current voice)	**hold E1**
PLAY (transport)	**B1**
MENU (save/load, settings)	**B2**
REC / pause / tap-tempo	**B3**
Clear RAM (fresh start)	**hold B1 at power-on**
Enter/exit **SINGLE**	**B1** (toggle)
Note Shift (in Single)	**E1 click @ row 16**
VELOCITY (velocity/prob/vol/condition)	**B2** to reach it → **E1/E2/E3/E4**

You want to...	Gesture
FILTER mode (per-voice DSP)	**B2 + B1, B2 first** → sliders on **E1–E4**
Sample Browser (set WAV)	**B1 + B2, B1 first / together** → confirm **E4 click**
Menu confirm / enter sub-menu	**E4 click** · back = **E1 click**
Recording **count-in**	**hold B3 > 300 ms**

> ⚠️ **The two combos differ only by order**: **B2 first** → FILTER mode; **B1 first (or both at once)** → Sample Browser. Wait ~350 ms between combos.

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20 · Common problems

Symptom	Fix
🔇 **I don't hear anything**	check the volume/channel volume (OLED), that the voice isn't muted (**hold E1**), and that the sample exists on the SD in the right format
▶️ **It won't start**	Play is **B1** (or **E3 click**); Pause during play is **B3**
🔄 **A knob goes the wrong way**	it's a CLK/DT swap from the wiring stage (see the build manual)
🚫 **The samples won't play**	wrong SD path/structure, or WAV not mono/16-bit/44.1 kHz → use `wavmaker.py`
🤔 **A combo opened the wrong mode**	mind the order: **B2 first** = FILTER, **B1 first** = Sample Browser; wait ~350 ms and retry
💾 **I want a fresh start**	**hold B1 at power-on** to clear the RAM; save songs to the SD from the Menu (B2)
⏱️ **Sample length**	long samples with continuous looping are supported

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Have fun! 🎶

*ichosynth runs **TŒRN** by SP_ (soundpauli) · toern.live · on a Teensy 4.1 with 4 KY-040 encoders, 3 tact switches and an SSD1306 OLED.*